Problem 89. Find all ordered pairs of positive integers (m, n) such that

$$m^3 + n^3 = m^2 + 42mn + n^2.$$

Solution. The answers are $(m, n) = (1, 7), (7, 1), (22, 22)$.

Let $d = \gcd(m, n)$. Then $m = ad$ and $n = bd$ where a, b, d are positive integers and $\gcd(a, b) = 1$. So we have

$$\begin{aligned} m^3 + n^3 = m^2 + 42mn + n^2 &\iff d^3\left(a^3 + b^3\right) = d^2\left(a^2 + 42ab + b^2\right) \\ &\iff d(a+b)\left(a^2 - ab + b^2\right) = a^2 + 42ab + b^2 \\ &\iff (da + db - 1)\left(a^2 - ab + b^2\right) = 43ab. \end{aligned}$$

Note that $\gcd(a^2 - ab + b^2, a) = \gcd(b^2, a) = 1$ and similarly $\gcd(a^2 - ab + b^2, b) = 1$. Therefore, we must have $a^2 - ab + b^2 \mid 43$. Thus, $a^2 - ab + b^2 = 1$ or $a^2 - ab + b^2 = 43$.

If $a^2 - ab + b^2 = 1$, then $(a - b)^2 + ab = 1$. As a, b are positive integers, the only possibility is $a = b = 1$. Then we need $2d = 44$, which gives $d = 22$. Thus, $x = y = 22$.

If $a^2 - ab + b^2 = 43$, without loss of generality we assume $a \geqslant b$. Then $43 = a^2 - ab + b^2 \geqslant ab \geqslant b^2$. So it suffices to check $b = 1, 2, 3, 4, 5, 6$. We can substitute each b in $a^2 - ab + b^2 = 43$ and solve for a. The only solutions are $(b, a) = (1, 7)$ and $(6, 7)$. When $(b, a) = (1, 7)$, we need $8d - 1 = 7$, i.e. $d = 1$. Thus, $(x, y) = (7, 1)$. By symmetry, $(x, y) = (1, 7)$ is another solution. When $(b, a) = (6, 7)$, we need $13d - 1 = 42$, which has no solution.

It is easy to check that all of $(x, y) = (1, 7), (7, 1), (22, 22)$ are solutions. So these are the only solutions.

□

Problem 90. Find all injective functions $f\colon \mathbb{R} \to \mathbb{R}$ such that for every real number x and every positive integer n,

$$\left|\sum_{m=1}^{n} m\left(f(x+m+1) - f(f(x+m))\right)\right| < 2020.$$

Solution.

Let $g(x) = f(x+1) - f(f(x))$. The condition becomes

$$\left|\sum_{m=1}^{n} mg(x+m)\right| < 2020$$

for any real number x and positive integer n. Note that

$$|(n+1)g(x+n+1)| = \left|\sum_{m=1}^{n+1} mg(x+m) - \sum_{m=1}^{n} mg(x+m)\right| \leqslant \left|\sum_{m=1}^{n+1} mg(x+m)\right| + \left|\sum_{m=1}^{n} mg(x+m)\right| < 2020 + 2020 = 4040.$$

This implies $|g(x+n+1)| < \dfrac{4040}{n+1}$ for any real number x and positive integer n. For any real number y and any positive integer n, we put $x = y - n - 1$ so that $|g(y)| < \dfrac{4040}{n+1}$. As this holds for any positive integer n, we must have $g(y) = 0$. This shows $f(y+1) = f(f(y))$ for any real number y. As f is injective, this yields $f(y) = y + 1$.

When $f(y) = y + 1$, the summation in the given inequality is 0, and so the inequality holds. Therefore, $f(y) = y + 1$ for any real number y is the only solution.

□

Problem 91. Let $\mathbb{R}^2 = \{(x, y)\colon x, y \in \mathbb{R}\}$. A *lattice point* is a point with integer coordinates and a *jump vector* is the vector produced after a jump.

A grasshopper is sitting on a lattice point in $\mathbb{R}^2$. Each second it jumps to another lattice point in such a way that the jump vector is constant. A hunter that knows neither the starting point of the grasshopper nor the jump vector (but knows that the jump vector for each second is constant) wants to catch the grasshopper. Each second the hunter can choose one lattice point in $\mathbb{R}^2$ and, if the grasshopper is there, he catches it. Can the hunter always catch the grasshopper in a finite amount of time?

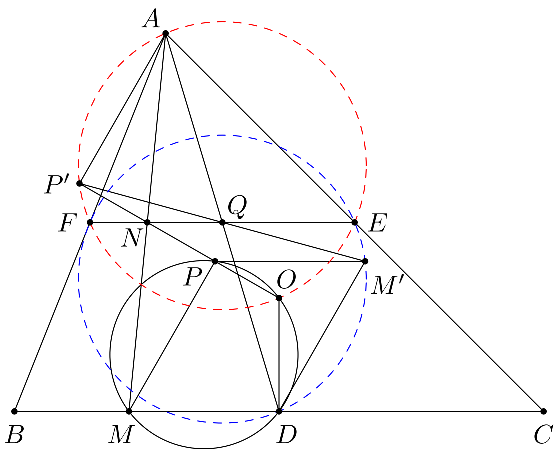
Solution.

Yes. Let (a_0, b_0) be the starting point of the grasshopper, and let (c_0, d_0) be the jump vector. Then the position of the grasshopper at time t is $(a_0 + c_0t, b_0 + d_0t)$. Since $\mathbb{Z}$ is countably infinite, so is $\mathbb{Z}^4$. Therefore, there exists a bijection $f : \mathbb{N} \to \mathbb{Z}^4$. A strategy of the hunter is to choose the lattice point $(a + ct, b + dt)$ at time $t \in \mathbb{N}$, where $(a, b, c, d) = f(t)$. From above, the grasshopper will be caught at time t_0, where $f(t_0) = (a_0, b_0, c_0, d_0)$. Such a t_0 exists because f is bijective.

One possible bijection $f : \mathbb{N} \to \mathbb{Z}^4$ can be constructed as follows. Any such bijection is equivalent to a labelling of the elements in $\mathbb{Z}^4$ by positive integers. We first label the only tuple (a, b, c, d) with $|a| + |b| + |c| + |d| = 0$ (i.e. $(0, 0, 0, 0)$) by 1. Then we label those tuples (a, b, c, d) with $|a| + |b| + |c| + |d| = 1$ by $2, 3, \ldots, 9$, etc. Since for each nonnegative integer k, there is a finite number of tuples (a, b, c, d) with $|a| + |b| + |c| + |d| = k$, we can label all tuples in this way.

□

Problem 92. Let ABC be acute-angled triangle with circumcentre O, and let D, E and F be the midpoints of BC, CA and AB, respectively. Let $M \neq D$ be an arbitrary point on BC. Suppose that the straight lines AM and EF intersect at point N and that the straight line ON cuts again the circumcircle of $\triangle ODM$ at point P. Prove that the reflection of point M with respect to the midpoint of DP lies on the nine-point circle of $\triangle ABC$.



Solution.

Let M' be the reflection of point M with respect to the midpoint of DP. Then $MDM'P$ is a parallelogram. Since F and E are the midpoints of AB and AC respectively, N is the midpoint of AM. Let Q be the intersection of AD and EF, which is the midpoint of AD as well as the midpoint of EF. Let P' be the reflection of point P with respect to N.

Since N is the midpoint of both AM and PP', $AP'MP$ is a parallelogram. Therefore, we have $\angle AP'O = \angle P'PM = \angle ODM = 90°$. Also, as $\angle AFO = 90° = \angle AEO$, the points A, P', F, O, E are concyclic.

Next, by the midpoint theorem, we have $NQ = \frac{1}{2}MD = \frac{1}{2}PM'$. As NQ is parallel to MD and hence to PM', and N is the midpoint of $P'P$, Q is the midpoint of $P'M'$. Now, reflect the points A, P', F, E with respect to the point Q. The images are D, M', E, F respectively. Since A, P', F, E are concyclic, so are D, M', E, F. This means M' lies on the nine-point circle (DEF) of $\triangle ABC$.

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